



Supersymmetry Dataset

DSc Endsem Project Presentation

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Dataset Description



- The Supersymmetry (SUSY) dataset supports binary classification of particle physics collision events based on the collected features of the collisions. (Domain : Physics)
- The dataset consists of 5 Million data points which exist in an 18 dimensional space.
- It consists of 18 input features and 1 label (Value is binary i.e. 0 or 1)
- The features are : lepton 1 pT, lepton 1 eta, lepton 1 phi, lepton 2 pT, lepton 2 eta, lepton 2 phi, missing energy magnitude, missing energy phi, MET_rel, axial MET, M_R, M_TR_2, R, MT2, S_R, M_Delta_R, dPhi_r_b, cos(theta_r1). These are kinematic features which all in some sense quantify and measure the movement of the particles.
- The initial eight are low level features and the rest are high level which were experimentally derived from the low level features.
- All attributes in the dataset are continuous with floating point values.
- There are no missing values in the dataset.

Problem Statement



Collisions at high energy particle colliders are a traditionally fruitful source of exotic particle discoveries. Finding these rare particles require solving difficult signal versus background classification problems, hence machine learning approaches are often used.

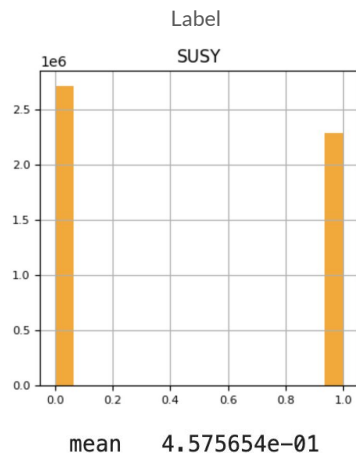
The primary tools for experimentation are particle accelerators which collide protons and/or antiprotons to create exotic particles (such as the Higgs Boson) that occur at extremely high-energy densities. Observing these particles and measuring their properties may yield critical insights about the very nature of matter.

The problem at hand is a binary classification problem in which the final prediction is that whether a specific kinematic configuration of colliding particles result in a supersymmetric event (Exotic particle is produced) or not. A label value of 0 implies that the collision was a background process and a label value of 1 implies that the collision was a supersymmetric event in which a new particle was generated.

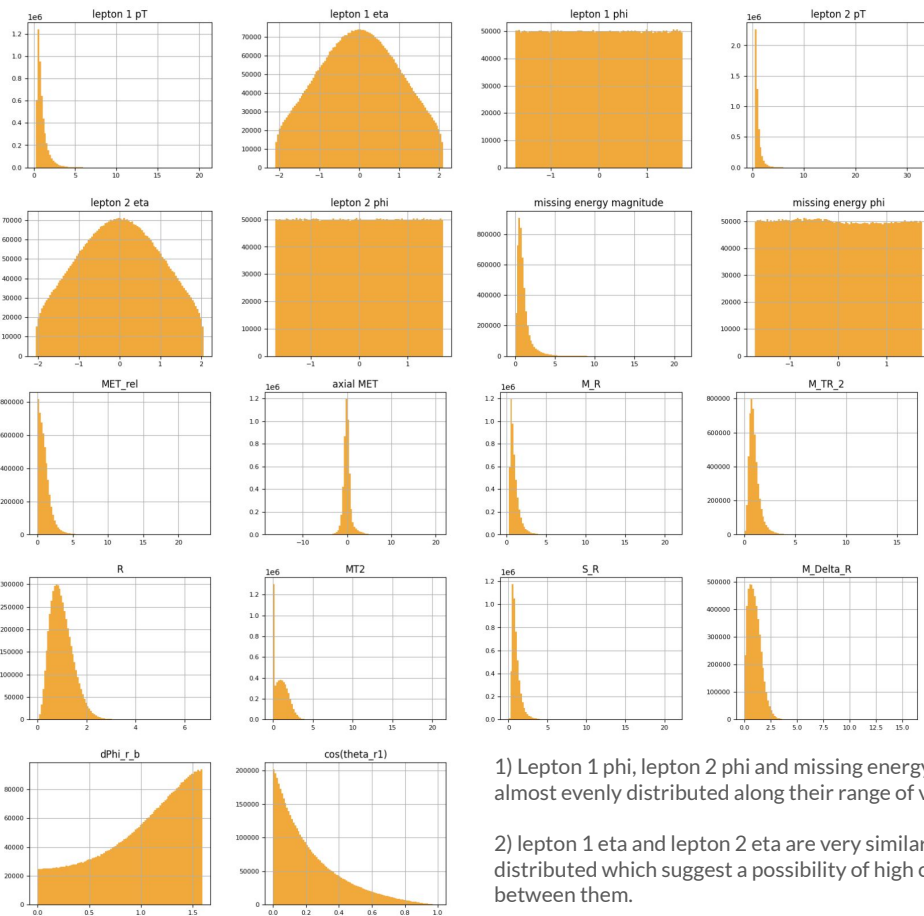
The conduction of such experiments tends to be very expensive with the requirement of skilled personnel and modern particle accelerators. Testing all possible combinations of kinematic configurations becomes infeasible in this case, so we rely on machine learning and analytical methods to try and mathematically predict the outcome of the experiment by keeping the general trends and observations from past experiments in mind.

EDA

1) Attribute Distributions



The number of 0's and 1's are almost identical and the mean is approximately 0.46 which indicates that there is no class imbalance in the data.



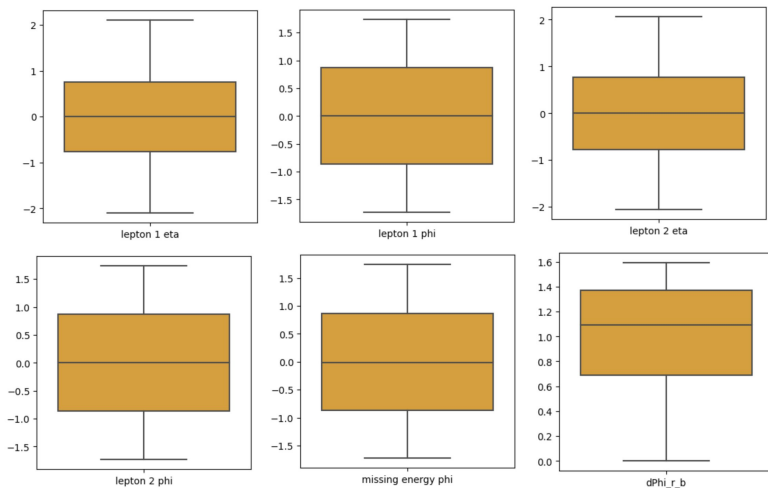
1) Lepton 1 phi, lepton 2 phi and missing energy phi are almost evenly distributed along their range of values.

2) lepton 1 eta and lepton 2 eta are very similarly distributed which suggest a possibility of high correlation between them.

EDA

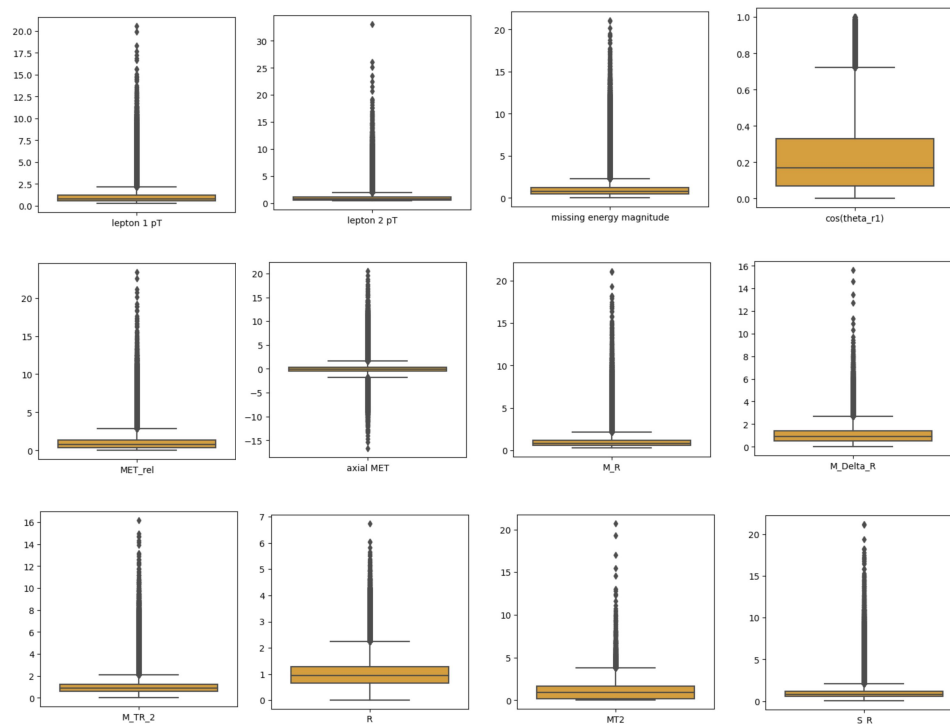
2) Checking for Outliers

Attributes With No Outliers



Some attributes are outlier free while others are not, these will be rectified in the preprocessing stage

Attributes With Outliers



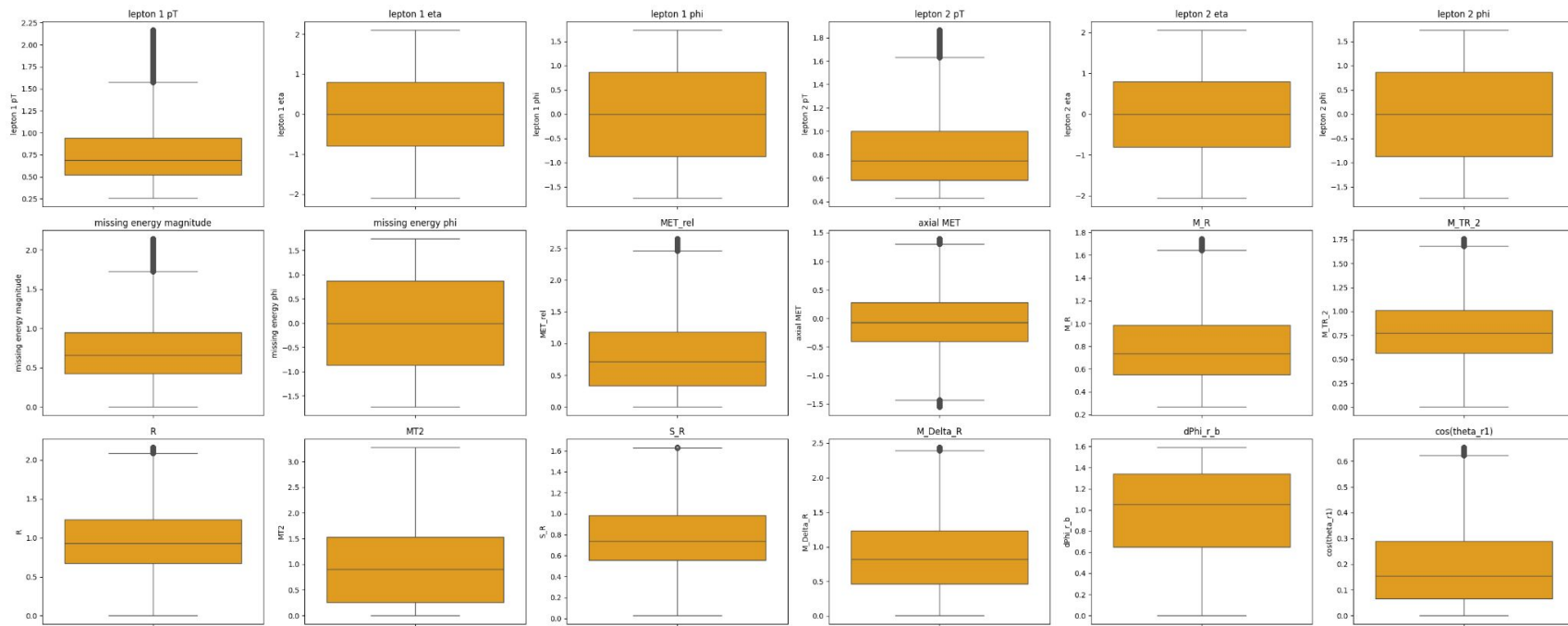
Preprocessing



- There were no missing values in dataset so no deletion or imputation were needed.
- All values in the dataset were between the range +2.5 and -1.5 (after outlier removal) so no scaling was needed as there were no values with high magnitude that would significantly affect the results.
- After observing the boxplots we determined there are a lot of outliers in the dataset so we decided to remove them.
- The Interquartile Range (IQR) method was applied to remove outliers. For each selected column, values outside 1.5 times the IQR from the first (Q1) and third quartiles (Q3) were considered outliers.
- After removing outliers the size of dataset was reduced from 5 million instances to 3,689,288 instances.

Box Plots

After Removing Outliers



Revamped feature selection

a) Conducted ANOVA tests to determine if the mean of a feature varies across the two classes (0 and 1) to determine if it has any predictive power. Removed all those features with an ANOVA score of less than 3000.

b) Conducted hypothesis tests for Spearman's correlation coefficient to determine monotonic relationships (both linear and nonlinear) between features to reduce redundant features. (critical p value was 0.05)

Null Hypothesis -> No monotonic correlation i.e. spearman's coefficient = 0

Alternate Hypothesis -> monotonic correlation i.e. spearman's coefficient != 0

- 1) M_R and S_R had a very high spearman coefficient of 0.91 and M_R had a higher spearman coefficient with the label (0.28 as compared to 0.26) hence we can drop S_R to reduce feature redundancy.
- 2) Similarly MT2 and M_Delta_R had a very high spearman coefficient of 0.92 and M_Delta_R had a higher spearman coefficient with the label (0.14 as compared to 0.04) hence we can drop MT2.

c) The final 10 features after feature elimination are shown on the right.

Feature Scores:

	Feature	ANOVA Score
0	lepton 1 pT	675192.006371
11	M_TR_2	441036.268108
6	missing energy magnitude	417632.828225
10	M_R	296913.882644
14	S_R	238314.409050
17	cos(theta_r1)	132758.035243
3	lepton 2 pT	124006.157720
15	M_Delta_R	102790.837225
8	MET_rel	68590.634611
13	MT2	18839.826415
16	dPhi_r_b	8411.773234
12	R	3079.101937
9	axial MET	77.198380
2	lepton 1 phi	5.178125
4	lepton 2 eta	1.520912
7	missing energy phi	1.261077
1	lepton 1 eta	0.006226
5	lepton 2 phi	0.002606

Data columns (total 10 columns):

#	Column	Dtype
0	lepton 1 pT	float64
1	lepton 2 pT	float64
2	missing energy magnitude	float64
3	MET_rel	float64
4	M_R	float64
5	M_TR_2	float64
6	R	float64
7	M_Delta_R	float64
8	dPhi_r_b	float64
9	cos(theta_r1)	float64

dtypes: float64(10)

memory usage: 309.6 MB

Models Employed



The models and their hyperparameters that were used for the purpose of binary classification were :

- 1) **Logistic Regression**
- 2) **Random Forests** (n_estimators=50)
- 3) **Support Vector Machine** (Kernelized) (kernel='rbf', max_iter=250)
- 4) **Perceptron** (max_iter=2000, tol=1e-3)

The training times and accuracies for all these models on raw data (no feature selection or dimensionality reduction) are summarized in the table below :

Model	Training Time (seconds)	Accuracy (Test)	Recall (Test)	Precision (Test)	F1 score (Test)
Logistic	14.55	76.6%	0.77	0.76	0.75
Random Forest	285.8	77.3%	0.77	0.77	0.76
SVM	484.82	62.4%	0.62	0.57	0.57
Perceptron	2.15	66.0%	0.66	0.63	0.59

Models Employed



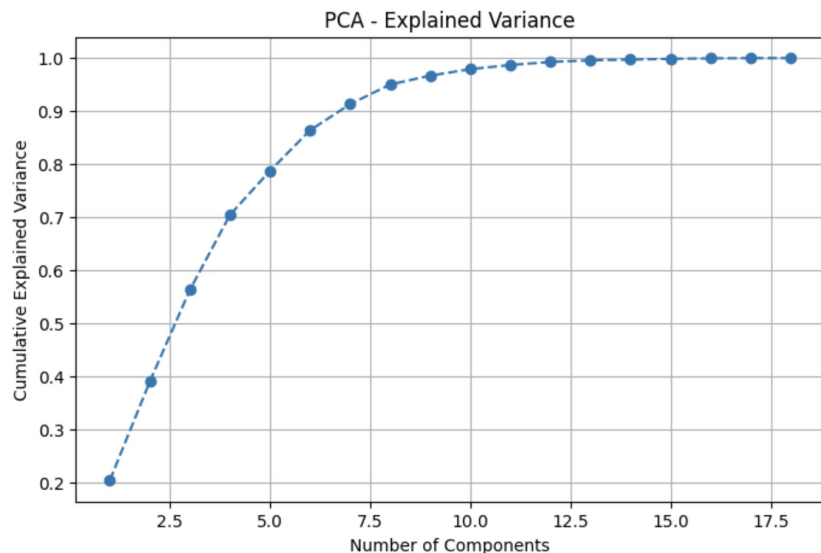
The metrics for each model after feature elimination (total 10 features and no scaling methods used) are given below :

Model	Training Time (seconds)	Accuracy (Test)	Recall (Test)	Precision (Test)	F1 score (Test)
Logistic	6.08	76.4%	0.76	0.76	0.75
Random Forest	206.45	76.3%	0.76	0.76	0.76
SVM	354.28	54.9%	0.55	0.51	0.53
Perceptron	2.04	68.0%	0.68	0.76	0.57

Next we will take a look at methods which aid in dimensionality reduction as they remove noise and redundancy in data, speed up the training phase and reduce overfitting (if any).

Dimensionality Reduction (PCA)

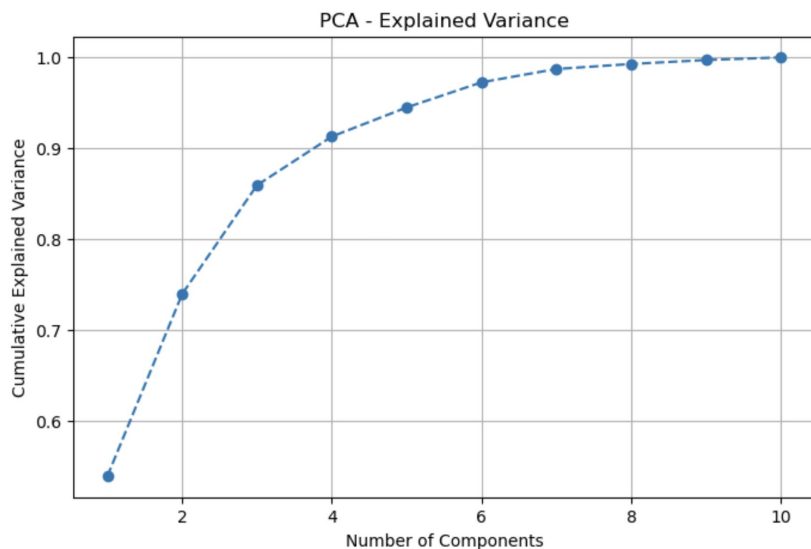
It was observed that only 8 components were able to explain more than 95% of variance in the data, we transformed the original data (no feature selection) and reduced the number of dimensions to 8 and trained the models. The statistics of which are on the right.



Model	Training Time (seconds)	Accuracy (Test)	Recall (Test)	Precision (Test)	F1 score (Test)
Logistic	1.11	69.9%	0.70	0.68	0.67
Random Forest	164.87	74.8%	0.75	0.74	0.74
SVM	339.86	61.2%	0.61	0.54	0.55
Perceptron	1.63	56%	0.56	0.54	0.55

Dimensionality Reduction (PCA+Feature Selection)

It was observed that only 6 components were able to explain more than 95% of variance in the data, we transformed the data after feature selection (10 total features), reduced the number of dimensions to 6 and trained the models. The statistics of which are on the right.



Model	Training Time (seconds)	Accuracy (Test)	Recall (Test)	Precision (Test)	F1 score (Test)
Logistic	1.41	74.3%	0.74	0.74	0.73
Random Forest	145.66	75.3%	0.75	0.75	0.74
SVM	182.95	66.2%	0.66	0.67	0.54
Perceptron	1.56	55.0%	0.55	0.55	0.55

Dimensionality Reduction (JL Matrix)



Considering the original data with no feature elimination (18 features).

Assume that the original data matrix (D) was of size $n \times d$ then the JL matrix that was created was of size $d \times d/2$ to effectively reduce the dimensions in half.

The original data matrix was of size $\sim 3.7M \times 18$ so the JL matrix that was chosen was of size 18×9 with elements sampled from a normal distribution $N(0, 1/18)$.

The matrix D was multiplied by the JL matrix to get the transformed data. The metrics for the models trained on this transformed data are given in the table on the right.

Model	Training Time (seconds)	Accuracy (Test)	Recall (Test)	Precision (Test)	F1 score (Test)
Logistic	2.86	70.3%	0.70	0.69	0.68
Random Forest	197.60	74.5%	0.75	0.74	0.73
SVM	250.79	61.0%	0.61	0.56	0.57
Perceptron	2.03	67.0%	0.67	0.70	0.55

Dimensionality Reduction (JL Mat+Feature Selection)

Considering the data after feature elimination (10 features).

Assume that the data matrix (D) was of size $n \times d$ then the JL matrix that was created was of size $d \times d/2$ to effectively reduce the dimensions in half.

The data matrix was of size $\sim 3.7M \times 10$ so the JL matrix that was chosen was of size 10×5 with elements sampled from a normal distribution $N(0, 1/10)$.

The matrix D was multiplied by the JL matrix to get the transformed data. The metrics for the models trained on this transformed data are given in the table on the right.

Model	Training Time (seconds)	Accuracy (Test)	Recall (Test)	Precision (Test)	F1 score (Test)
Logistic	2.07	75.1%	0.75	0.75	0.74
Random Forest	133.98	75.04%	0.75	0.74	0.74
SVM	214.75	46.7%	0.47	0.70	0.42
Perceptron	1.59	68.0%	0.68	0.71	0.58

Conclusion and Future Work



- 1) In this project we used two techniques for reducing the dimensions (compressing the information in the data) namely PCA and dimensionality reduction using JL matrices and employed 4 different models for binary classification.
- 2) In general it was observed that the training time reduced significantly for all models upon applying these scaling techniques with only minimal loss in accuracy and other performance metrics except in a very few cases.
- 3) Further work may include the following :
 - a) Trying out more models for binary classification.
 - b) Varying the hyperparameters and testing further.
 - c) Experimenting with other dimensionality reduction/scaling techniques.